

Balloon Management System

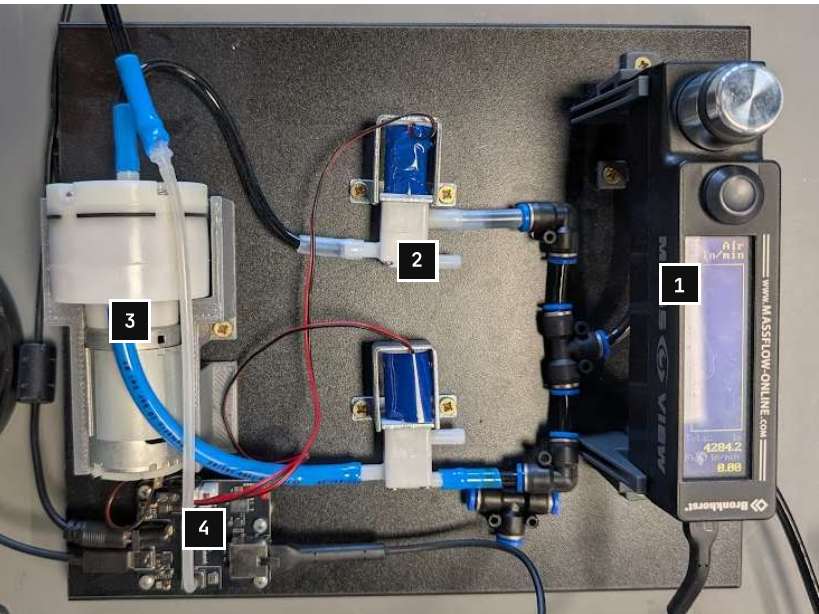
Programmable pneumatic platform for balloon preparation and pressure-system testing

MY ROLE SYSTEM REQUIREMENTS • COMPONENT SELECTION • PNEUMATIC INTEGRATION • CONTROL SOFTWARE • VALIDATION

Early flights revealed premature failures in some superpressure balloons. Pre-stretching the envelope at its simulated float pressure improved durability, so stretching became a standard step before flight. The target stretch pressure comes from balloon and flight simulation, not from an arbitrary choice.

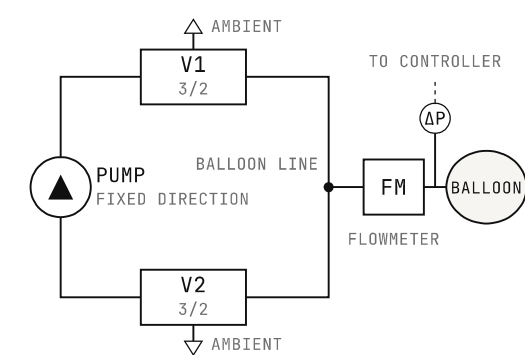
The procedure is easy to describe and impractical to run manually: inflate, reach a target differential pressure, hold it while the material relaxes at operating stress, then deflate. A single stretch can occupy an operator for hours of pressure watching and pump correction. I defined the requirements for an automated platform to run it, selected and integrated the hardware, and built the control software.

80–1100 L	13 sL/min	±2 Pa	50+
BALLOON VOLUME RANGE RUN ON THE PLATFORM	MAXIMUM PUMP FLOW	OBSERVED DIFFERENCE VS PRESSUREWELL REFERENCE	BALLOONS STRETCHED



- 1 Mass flowmeter in the gas line
- 2 Solenoid valve pair, commanded together
- 3 Diaphragm pump
- 4 Controller PCB and drivers

FIG. 1 — PNEUMATIC H-BRIDGE



INFLATE	V1 → balloon line · V2 → ambient
DEFLATE	V1 → ambient · V2 → balloon line
ISOLATE	Both valves → ambient. Balloon line sealed from the pump — not used

Fig. 1 — Pneumatic H-bridge. Two 3/2 valves swap the pump's discharge and suction between ambient and the balloon line, so a fixed-direction pump both inflates and deflates. The flowmeter sits in the balloon line and is read during inflation.

REQUIREMENT I DEFINED

Controlled gas delivery	Diaphragm pump, 13 sL/min maximum, commanded 0–255
Flow measurement across several gases	Bronkhorst MASS-VIEW, RS-485 Modbus RTU
Differential pressure sensing at the balloon	On-board sensor, cross-checked against a Pressurewell reference
Electrically switched pneumatic routing	Solenoid valves driven directly from the controller board
Reversible flow path for inflation and deflation	Two 3/2 valves swap the pump's suction and discharge between ambient and the balloon line
Programmable, unattended sequence control	Python/PyQt test applications over serial
Data acquisition and test records	Per-test CSV, event log, metadata, saved plots

HARDWARE DEVELOPMENT

The custom ESP32 controller PCB was designed by the project's electronics engineer, who also began the embedded firmware. I integrated the board into the pneumatic system, extended the firmware and interface, and defined the serial commands my Python test applications rely on.



ESP32 control board

Why the pump could not simply be switched on

From the controller's point of view a balloon is a strongly nonlinear load. Early in inflation the envelope accepts a large volume for almost no pressure rise. As it approaches its inflated geometry the same added volume produces a steep pressure rise, and during a long hold only small corrections are needed.

A single pump command cannot serve all three regimes: fast enough to fill a 1100 L envelope in reasonable time is far too fast to approach a target without overshoot. Several iterations of the control software were needed before the sequence ran unattended.

STAGED CONTROL STRATEGY

01	02	03	04
Fast fill	Controlled approach	Pressure hold	Auto-deflate
Maximum pump command until a configurable switch pressure	Pump command trimmed against a flow target, not a pressure target	Pump commanded only when averaged pressure leaves a symmetric dead band	Valve pair reversed, pump run until the deflation condition is met
OPEN LOOP	CLOSED LOOP ON FLOW	DEAD-BAND REGULATION	TERMINATED ON PRESSURE

The approach phase regulates the rate at which gas enters the envelope rather than the pressure itself, which keeps the fill controllable as the pressure response changes.

FIG. 3 – MEASURED STRETCH TRACE

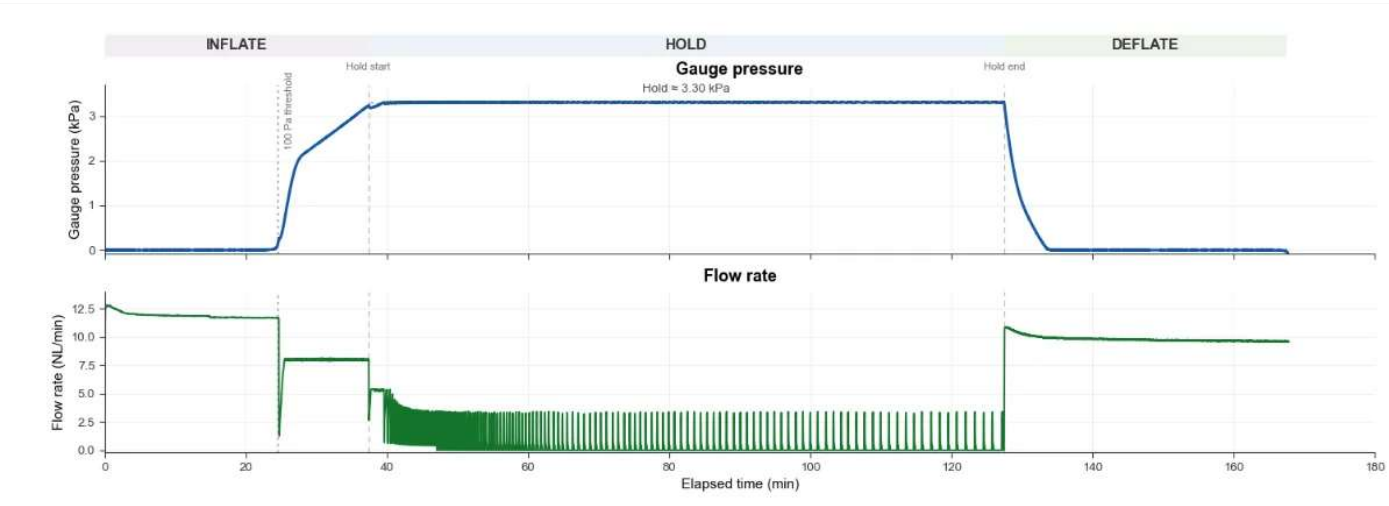


Fig. 3 – Logged stretch. Phase transitions are visible in the pressure slope, with the 3.3kPa target pressure and the ≈100 Pa crossing marked. Values are read from a logged run, not estimated.

PRACTICAL INFLATED-VOLUME ESTIMATE

A superpressure envelope accepts gas at almost no differential pressure until it approaches its free volume. The software integrates measured flow throughout inflation and records the accumulated volume at the first crossing of approximately 100 Pa — the practical threshold used in this workflow, not a universal definition of free volume.

INFLATE → ΔP ≈ 100 PA → VOLUME RECORDED

DESIGN DECISION – MULTI-GAS FLOW MEASUREMENT

Need: flow measurement for several gases used in balloon and pneumatic testing, including the He and Ar used when preparing helium–argon mixtures.

Problem: For He measurements Bronkhorst offers a helium-specific meter that would measure helium directly but would be useful for essentially one gas.

Decision: the multi-gas MASS-VIEW for the compatible gases (Air, Ar, N2, etc), with helium quantity determined separately by free-lift measurement on a scale. Versatility was worth more than one direct measurement.

SAFETY INTERLOCKS

- Overpressure → pump stopped, valves commanded to vent
- Pressure reference lost or stale → test aborted
- Controller link lost → test aborted
- Pressure fails to rise while the pump is commanded on → fault raised

Additional instrument-state interlocks are implemented for Pressurewell-based valve testing.

BALLOON STRETCHING → VOLUME / LEAK TESTING → PRV CHARACTERISATION

Built to automate one repetitive balloon preparation step, the same hardware became a reusable pressure and flow test platform. For valve characterisation it provides pump and valve actuation and flow acquisition while an independent Pressurewell supplies the pressure reference, deliberately keeping the device under test out of the control loop.

WHAT I WOULD CHANGE

Pressure regulation

The dead-band controller is adequate for stretching. A proportional or PI controller would give smoother corrections and tighter regulation during the hold.

Pneumatic architecture

The two 3/2 valves are always commanded as a pair, so a single 5/2 valve would do the same job with one output. The unused fourth combination — both to ambient — would seal the balloon line from the pump for a cleaner hold-phase leak measurement.